



ENERGY FACILITY VISIT REPORT

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FOOD WASTE TREATMENT

BIOGAS PLANT

I visited the Biogas plant in IIT Bombay campus. This biogas plant is in front of Hostel-6 and provides biogas to Hostel-13.

This biogas plant uses waste food (largely, gathered from the mess) as a fuel to generate gas that is again used to cook food in the mess.

Plant details and usage

Design parameters

1. Consists of different parts such as: -
 - a. Shredder.



b. Mixing tank.



c. Hydrolysis tank.

d. Digester.



e. Dewatering system.



f. Scrubber.



g. Compressor.

h. Filters.



i. Storage.



2. Total cost of plant – Rs. 75,97,980.
3. Operating and maintenance cost - Rs. 9,42,642.00 per year.
4. The fuel mostly consists of food waste.

Operating parameters

1. Capacity – 2 TPD.
2. A single digester can handle a load of up to 2000kgs.
3. Hydrolysis tank can handle a load of up to 1000kgs.

4. For every 50kgs of waste organic material 10 litres of water is added.
5. Everyday around 1430kgs of waste food is brought to the plant.
6. Temperature of about 35-37° C at mesophilic condition required.
7. pH of about 6.8-7.2 required.
8. C/N ratio should be between 25:1 – 30:1.

Performance parameters

1. Very environment friendly, transforms waste into energy.
Also, produces manure.
2. Requires to be set up in an isolated area with very less population due to the issue of smell.
3. It produces very low-pressure flame which can be only used to boil water and pulses.

4. Average calorific value of biogas = $20\text{MJ} / \text{m}^3$.
5. Average calorific value of fuel = $23.25\text{MJ} / \text{kg}$.
6. Efficiency is very low.
7. Cannot be used on very large scale.
By using biofuel, cooking requires more time than by using LPG.

Energy Calculations

Total mass input = 1430kgs (Solid waste food) + 28.6 (Water added) = 1458.6kgs

Total mass output = 45 kgs

Density of biogas = 1.25 kg/m^3

Volume of biogas used = $45 / 1.25 = 36\text{ m}^3$

Total input energy = $1458.6 \times 23.25 = 33912.45\text{ MJ}$

Total output energy = $36 \times 20 = 720\text{ MJ}$

Efficiency = Total output energy / Total
input energy = $720 / 33912.45 = 0.021 =$
2.1%

P.T.O.

ANNEXURE



Photo taken in front of biogas plant.

REFERENCES

<https://web.iitd.ac.in/~vkvijay/files/Biogas%20technology.pdf>.